

AN ELEMENTARY
INTRODUCTION TO
MICROECONOMICS



KUTUSH KUMAR, KUTUM PAKHI

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reasoning should use the facts given in the question and keep timing and units consistent.

15.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

15.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

15.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XVI · LEASING

16.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

16.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XI · REQUIRED RETURN, CAPM AND MARKET EFFICIENCY

11.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

11.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

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11.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

11.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XII · VALUATION OF BONDS

12.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

12.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XIII · VALUATION OF SHARES

13.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

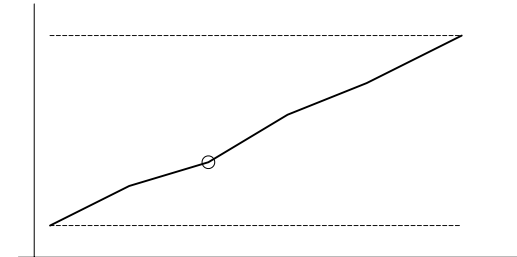


FIG. 1.1. What Finance Is figure 1

1.1 FINANCE IN PLAIN WORDS

This section builds the idea of finance in plain words before moving to application.

Finance. The activity of finding money, using it, and judging whether the result was worth it.

The term finance gives the chapter a precise name for this part of the problem.

Cash flow. Money that actually moves into or out of a business.

The term cash flow gives the chapter a precise name for this part of the problem.

Profit. What a business earns from sales after subtracting the costs of earning them.

The term profit gives the chapter a precise name for this part of the problem.

The central question in finance in plain words is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The purpose here is to make finance in plain words concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Finance** means The activity of finding money, using it, and judging whether the result was worth it. A shop owner who borrows to buy stock does finance. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash flow** means Money that actually moves into or out of a business. Paying a supplier Tk 5,000 is a cash flow of minus Tk 5,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Profit** means What a business earns from sales after subtracting the costs of earning them. Sales of Tk 100 and costs of Tk 80 leave profit of Tk 20. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

reasoning should use the facts given in the question and keep timing and units consistent.

06.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

06.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

06.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

06.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

06.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER VII · CAPITAL BUDGETING: PROCESS AND CASH FLOWS

07.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

07.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

07.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER VIII · EVALUATING PROJECTS: PAYBACK AND NET PRESENT VALUE

08.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

08.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

1.3 THE FINANCE FUNCTION INSIDE A FIRM

This section builds the idea of the finance function inside a firm before moving to application.

Company. A business set up as a legal person of its own, owned through shares, so the owners are separate from the business..

The term company gives the chapter a precise name for this part of the problem.

The central question in the finance function inside a firm is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

The purpose here is to make the finance function inside a firm concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Company** means A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the finance function inside a firm, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to what finance is. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

ANSWERS TO QUESTIONS

CHAPTER I · WHAT FINANCE IS

01.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

01.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

01.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

01.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

01.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER II · THE FINANCIAL MANAGER'S DECISIONS

02.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

02.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

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02.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

02.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER III · THE GOAL OF THE FIRM AND AGENCY THEORY

03.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

03.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

A short restatement keeps the financial meaning attached to the calculation.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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Underwriter. An institution that promises to buy any shares of a new issue that the public does not. (*Chapter XV*)

Unsystematic risk. Risk that affects one firm or industry only; diversification can remove it. (*Chapter XI*)

Valuation. Finding what an asset is worth today by adding the present values of the cash it will bring. (*Chapter XII*)

Variable cost. A cost that rises and falls with the number of units sold, such as the price paid for each book. (*Chapter X*)

Variance. The probability-weighted average of the squared gaps between each result and the expected value. (*Chapter IX*)

Wealth maximisation. Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. (*Chapter III*)

Weighted average cost of capital. The average cost of a firm's different sources of money, each weighted by its share of the total. (*Chapter XV*)

Working capital. The money a business has tied up in everyday trading: cash, stock and money owed to it. (*Chapter XIV*)

Yield to maturity. The yearly return an investor earns by buying a bond at today's price and holding it to maturity. (*Chapter XII*)

REVIEW QUESTIONS

01.T1 say in plain words what finance covers and how it differs from accounting and from management

01.T2 tell profit from cash flow using a new set of sales and collection figures

01.T3 compare sole proprietorship, partnership and company for a new tailoring shop

PAUSE AND CHECK 01.P1 After the finance definition, state the rule in your own words.

PAUSE AND CHECK 01.P2 After profit versus cash flow, state the rule in your own words.

percentage change in EBIT divided by the percentage change in sales. (*Chapter X*)

Opportunity cost. What you give up by choosing one use of money instead of the best alternative. (*Chapter IV*)

Ordinary annuity. An annuity whose payments fall at the end of each period. (*Chapter VI*)

Partnership. A business owned by two or more people who agree to run it together and share its profit. (*Chapter I*)

Payback period. The time a project takes to return its initial investment from its cash flows. (*Chapter VIII*)

Perpetuity. A series of equal payments that continues for ever. (*Chapter XIII*)

Portfolio. A collection of investments held together by one investor. (*Chapter XI*)

Preferred stock. Shares that pay a fixed dividend, which must be paid before any dividend on common stock. (*Chapter X*)

Premium bond. A bond that sells for more than its face value. (*Chapter XII*)

Present value. What a sum due in the future is worth today. (*Chapter V*)

Principal. A person who hires another to act for them. A shop owner who hires a manager is a principal. (*Chapter III*)

Principal sum. The original amount lent, borrowed or deposited, before any interest is added. (*Chapter IV*)

Probability distribution. A list of possible results with the chance of each, the chances adding to one. (*Chapter IX*)

Profit. What a business earns from sales after subtracting the costs of earning them. Sales of Tk 100 and costs of Tk 80 leave profit of Tk 20. (*Chapter I*)

Profit maximisation. Aiming to make the largest profit possible, usually within one year. (*Chapter III*)

Profitability index. Present value of future cash flows divided by the initial investment. (*Chapter VIII*)

Rate of return. The return in one period divided by the sum invested, written as a percentage. (*Chapter IX*)

Relevant cash flow. A cash flow that changes because of the decision being studied. (*Chapter VII*)

Required return. The least return an investor will accept for taking a given risk. (*Chapter XI*)

Residual claim. The right to whatever is left after lenders and preferred owners have been paid. (*Chapter XIII*)

Retained earnings. Profit kept in the business instead of being paid out as dividends. (*Chapter XV*)

Return. The gain or loss from an investment, including income received and any change in value. (*Chapter IX*)

Rights issue. An offer of new shares to existing shareholders, in proportion to their holdings. (*Chapter XV*)

Risk. The chance that the actual result will differ from the result you expect. (*Chapter IX*)

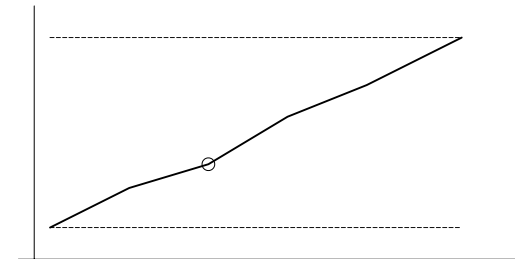


FIG. 2.1. *The Financial Manager's Decisions figure 1*

2.1 THE THREE BIG DECISIONS

This section builds the idea of the three big decisions before moving to application.

Financial manager. The person who decides how a business finds, uses and protects its money.

The term financial manager gives the chapter a precise name for this part of the problem.

Investment decision. A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money.

The term investment decision gives the chapter a precise name for this part of the problem.

Financing decision. A choice about where the money comes from: owners' own funds, a loan, or new owners.

The term financing decision gives the chapter a precise name for this part of the problem.

Dividend. A share of a company's profit paid out to its owners instead of being kept in the business.

The term dividend gives the chapter a precise name for this part of the problem.

The central question in the three big decisions is what changes when the underlying decision changes.

The purpose here is to make the three big decisions concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Financial manager** means The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Investment decision** means A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financing decision** means A choice about where the money comes from: owners' own funds, a loan, or new owners. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Dividend** means A share of a company's profit paid out to its owners instead of being kept in the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Dividend valuation model. A way of valuing a share as the present value of the dividends it is expected to pay. (*Chapter XIII*)

Earnings per share. Earnings left for owners of common stock divided by the number of shares. (*Chapter X*)

EBIT. Earnings before interest and tax: sales minus operating costs. (*Chapter X*)

Efficient market. A market in which prices already reflect the available information. (*Chapter XI*)

Equity. What is left for the owners when liabilities are subtracted from assets. (*Chapter II*)

Expected value. The average of possible results, each weighted by its probability. (*Chapter IX*)

Face value. The amount a bond repays at its end, for example Tk 1,000. (*Chapter XII*)

Factor table. A table of ready-made multipliers that saves repeating a long calculation. (*Chapter IV*)

Finance. The activity of finding money, using it, and judging whether the result was worth it. A shop owner who borrows to buy stock does finance. (*Chapter I*)

Financial lease. A long lease, not cancellable, that covers most of the asset's life and works like a loan. (*Chapter XVI*)

Financial leverage. The effect of fixed financing payments on how owners' earnings respond to EBIT. Measured by the degree of financial leverage (DFL). (*Chapter X*)

Financial manager. The person who decides how a business finds, uses and protects its money. In a small shop this may be the owner. (*Chapter II*)

Financial planning. Estimating the money a business will need and receive in future, and arranging how to meet the gap. (*Chapter II*)

Financial risk. Extra risk that comes from borrowing, because interest must be paid whatever the sales. (*Chapter IX*)

Financing decision. A choice about where the money comes from: owners' own funds, a loan, or new owners. (*Chapter II*)

Fixed cost. A cost that stays the same however many units are sold, such as shop rent. (*Chapter X*)

Float. Money tied up in payments that are sent but not yet usable, such as a cheque in the post. (*Chapter XIV*)

Future value. What a sum today will be worth at a later date when it earns interest. (*Chapter IV*)

Incentive. A reward, such as a bonus, that encourages a person to act in a chosen way. (*Chapter III*)

Independent project. A project that can be accepted or rejected without affecting any other project. (*Chapter VII*)

Initial investment. The cash flows at the start of a project to buy and prepare the asset. (*Chapter VII*)

Initial public offering. A company's first sale of shares to the public. (*Chapter XV*)

Instalment. One of a series of regular payments used to repay a loan or build a sum. (*Chapter VI*)

The business meaning matters as much as the arithmetic.

This section builds the idea of decisions on the balance sheet before moving to application.

Balance sheet. A statement listing what a business owns and what it owes on one date..

The term balance sheet gives the chapter a precise name for this part of the problem.

Asset. Something a business owns that has value, such as cash, stock or a van..

The term asset gives the chapter a precise name for this part of the problem.

Liability. Money a business owes to others, such as a bank loan or an unpaid supplier bill..

The term liability gives the chapter a precise name for this part of the problem.

Equity. What is left for the owners when liabilities are subtracted from assets..

The term equity gives the chapter a precise name for this part of the problem.

The central question in decisions on the balance sheet is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what a financial manager does, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

2.3 DECISIONS ON THE BALANCE SHEET

The purpose here is to make decisions on the balance sheet concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Balance sheet** means A statement listing what a business owns and what it owes on one date. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Asset** means Something a business owns that has value, such as cash, stock or a van. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Liability** means Money a business owes to others, such as a bank loan or an unpaid supplier bill. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Equity** means What is left for the owners when liabilities are subtracted from assets. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For decisions on the balance sheet, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a

GLOSSARY

Accounts payable. Money a business owes its suppliers for goods already received. (*Chapter XIV*)

Accounts receivable. Money that customers owe a business for goods already delivered. (*Chapter XIV*)

After-tax cash outflow. A payment's cost after subtracting the tax it saves. (*Chapter XVI*)

Agency cost. A cost the principal bears to reduce the agency problem or from the loss it still causes. (*Chapter III*)

Agency problem. A conflict that arises when an agent's own interests differ from those of the principal. (*Chapter III*)

Agent. A person who acts for a principal and makes decisions for the principal. A hired shop manager is an agent. (*Chapter III*)

Annuity. A series of equal payments made at equal intervals for a fixed number of periods. (*Chapter VI*)

Annuity due. An annuity whose payments fall at the start of each period. (*Chapter VI*)

Asset. Something a business owns that has value, such as cash, stock or a van. (*Chapter II*)

Balance sheet. A statement listing what a business owns and what it owes on one date. (*Chapter II*)

Beta. A number showing how strongly an investment's return moves with the market; 1.0 means it moves with the market. (*Chapter XI*)

Bond. A loan made to a company or government, repaid on a set date, with interest paid on the way. (*Chapter XII*)

Break-even point. The sales volume at which total revenue equals total cost, so profit is zero. (*Chapter X*)

Business risk. Risk that comes from the firm's own trade, such as falling sales or rising costs, before any borrowing. (*Chapter IX*)

Capital asset pricing model. A formula for required return: risk-free rate plus beta times the market risk premium. (*Chapter XI*)

Capital budgeting. Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. (*Chapter VII*)

Capital expenditure. Spending on an asset that will help a business for more than one year. (*Chapter VII*)

Capital gain. The profit from selling an asset for more than its purchase price. (*Chapter XIII*)

Capital market. The market in which long-term money is raised by selling bonds and shares. (*Chapter XV*)

Cash conversion cycle. The operating cycle minus the days a business takes to pay its suppliers. (*Chapter XIV*)

Cash discount. A price cut a supplier gives for paying early, such as 2% off if paid within 10 days. (*Chapter XIV*)

Cash flow. Money that actually moves into or out of a business. Paying a supplier Tk 5,000 is a cash flow of minus Tk 5,000. (*Chapter I*)

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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APPENDIX A

Maths You Will Use



This section builds the idea of profit maximisation and its limits before moving to application.

Profit maximisation. Aiming to make the largest profit possible, usually within one year..

The term profit maximisation gives the chapter a precise name for this part of the problem.

Wealth maximisation. Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is..

The term wealth maximisation gives the chapter a precise name for this part of the problem.

Shareholder. A person who owns part of a company by holding its shares..

The term shareholder gives the chapter a precise name for this part of the problem.

Share price. The price at which one share of a company can be bought or sold today..

The term share price gives the chapter a precise name for this part of the problem.

The central question in profit maximisation and its limits is what changes when the underlying decision changes.

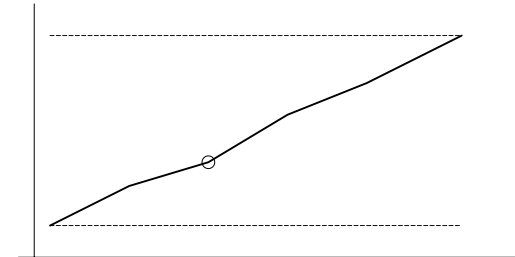


FIG. 3.1. *The Goal of the Firm and Agency Theory figure 1*

3.1 PROFIT MAXIMISATION AND ITS LIMITS

The purpose here is to make profit maximisation and its limits concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profit maximisation** means Aiming to make the largest profit possible, usually within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Wealth maximisation** means Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Shareholder** means A person who owns part of a company by holding its shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Share price** means The price at which one share of a company can be bought or sold today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

United States Leasing Corporation, founded in San Francisco in 1952, an early firm of the modern equipment-leasing industry. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

16.T1 say what a lease is and compare operating and financial leases with a new asset

16.T2 list the steps of the lease-or-buy comparison

16.T3 compute the after-tax present value of a new lease and purchase

16.T4 explain the tax shield of interest and depreciation

PAUSE AND CHECK 16.P1 After kinds of lease, state the rule in your own words.

PAUSE AND CHECK 16.P2 After after-tax outflows, state the rule in your own words.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

The business meaning matters as much as the arithmetic.

This section builds the idea of owners, managers and agency theory before moving to application.

Agency cost. A cost the principal bears to reduce the agency problem or from the loss it still causes..

The term agency cost gives the chapter a precise name for this part of the problem.

Incentive. A reward, such as a bonus, that encourages a person to act in a chosen way..

The term incentive gives the chapter a precise name for this part of the problem.

The central question in owners, managers and agency theory is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For wealth maximisation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

3.3 OWNERS, MANAGERS AND AGENCY THEORY

The purpose here is to make owners, managers and agency theory concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Agency cost** means A cost the principal bears to reduce the agency problem or from the loss it still causes. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Incentive** means A reward, such as a bonus, that encourages a person to act in a chosen way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For owners, managers and agency theory, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the goal of the firm and agency theory. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

AN ELEMENTARY INTRODUCTION TO MICROECONOMICS

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and

larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

16.2 KINDS OF LEASE

The purpose here is to make kinds of lease concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating lease** means A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial lease** means A long lease, not cancellable, that covers most of the asset's life and works like a loan. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sale-and-leaseback** means A deal in which a firm sells an asset it owns and leases it back from the buyer. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For kinds of lease, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

16.3 THE LEASE-OR-BUY DECISION

The purpose here is to make the lease-or-buy decision concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the

This section builds the idea of kinds of lease before moving to application.

Operating lease. A short lease, cancellable, where the lessor keeps most of the risks of owning the asset..

The term operating lease gives the chapter a precise name for this part of the problem.

Financial lease. A long lease, not cancellable, that covers most of the asset's life and works like a loan..

The term financial lease gives the chapter a precise name for this part of the problem.

Sale-and-leaseback. A deal in which a firm sells an asset it owns and leases it back from the buyer..

The term sale-and-leaseback gives the chapter a precise name for this part of the problem.

The central question in kinds of lease is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the lease-or-buy decision before moving to application.

SOURCES

Chapter 1, 'What Finance Is', section 1.1 'Finance in plain words' — profit as defined there; Chapter 5, 'Present Value and Discounting', section 5.1 'Discounting a single sum' — how timing is measured in wealth..

REVIEW QUESTIONS

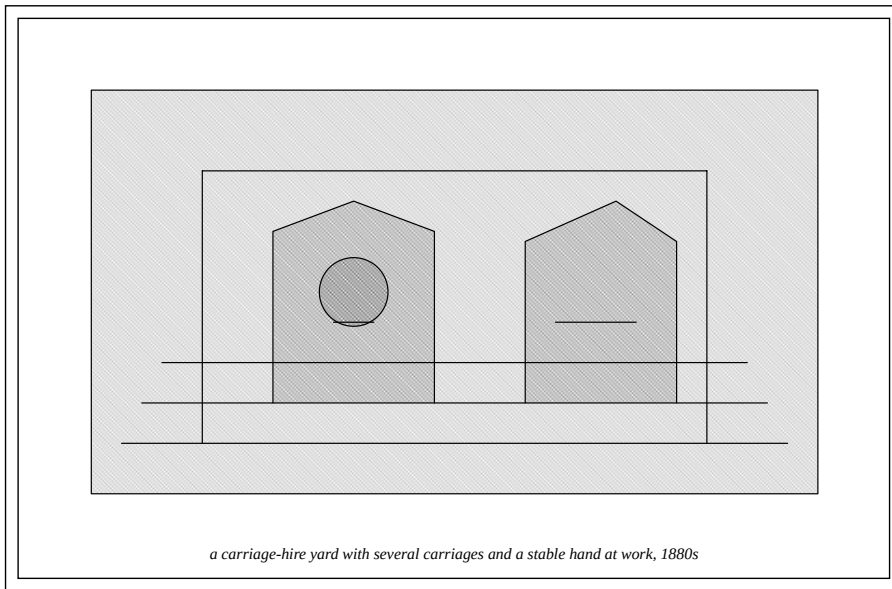
03.T1 explain why the aim of the largest profit can mislead, using new yearly figures

03.T2 define wealth maximisation and say how it differs from profit maximisation

03.T3 describe an agency problem and two remedies in a hospital-equipment firm

PAUSE AND CHECK 03.P1 After profit maximisation, state the rule in your own words.

PAUSE AND CHECK 03.P2 After the agency relationship, state the rule in your own words.



CHAPTER XVI

Leasing

A practical question about leasing



This chapter introduces leasing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of why time changes the value of money before moving to application.

Time value of money. The idea that a taka received today is worth more than a taka received later..

The term time value of money gives the chapter a precise name for this part of the problem.

Opportunity cost. What you give up by choosing one use of money instead of the best alternative..

The term opportunity cost gives the chapter a precise name for this part of the problem.

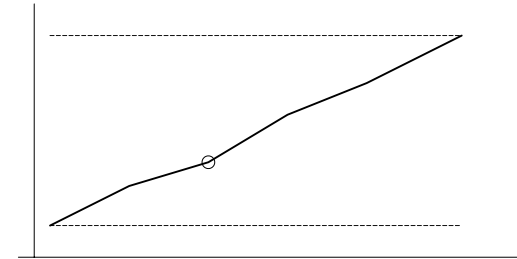


FIG. 4.1. Interest and Future Value figure 1

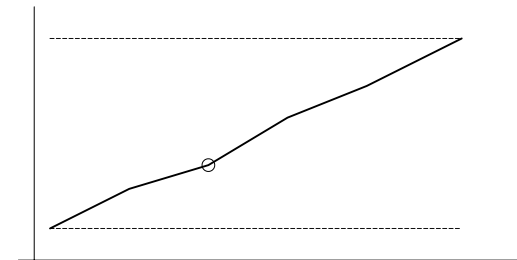


FIG. 4.2. Interest and Future Value figure 2

4.1 WHY TIME CHANGES THE VALUE OF MONEY

The purpose here is to make why time changes the value of money concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Time value of money** means The idea that a taka received today is worth more than a taka received later. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Opportunity cost** means What you give up by choosing one use of money instead of the best alternative. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Long-Term Financing and the Capital Market is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

Compound interest. Interest calculated on the principal sum and also on interest already earned..

The term compound interest gives the chapter a precise name for this part of the problem.

Future value. What a sum today will be worth at a later date when it earns interest..

The term future value gives the chapter a precise name for this part of the problem.

The central question in simple and compound interest is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of future value of a single sum before moving to application.

Compounding period. The length of time after which interest is added to the balance, such as a year or six months..

The term compounding period gives the chapter a precise name for this part of the problem.

Timeline. A line marking each period, with the cash flow written at the point where it occurs..

The term timeline gives the chapter a precise name for this part of the problem.

In this chapter, **Compound interest** means Interest calculated on the principal sum and also on interest already earned. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Future value** means What a sum today will be worth at a later date when it earns interest. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For simple and compound interest, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

4.3 FUTURE VALUE OF A SINGLE SUM

The purpose here is to make future value of a single sum concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Compounding period** means The length of time after which interest is added to the balance, such as a year or six months. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Timeline** means A line marking each period, with the cash flow written at the point where it occurs. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cost of capital** means The return a firm must earn on its investments to satisfy those who supply its money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Weighted average cost of capital** means The average cost of a firm's different sources of money, each weighted by its share of the total. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the cost of capital, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to long-term financing and the capital market. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

Cost of capital. The return a firm must earn on its investments to satisfy those who supply its money..

The term cost of capital gives the chapter a precise name for this part of the problem.

Weighted average cost of capital. The average cost of a firm's different sources of money, each weighted by its share of the total..

The term weighted average cost of capital gives the chapter a precise name for this part of the problem.

The central question in the cost of capital is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

A calculation can be correct and still answer the wrong question. For term loans and the features of financing, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

15.2 DEBT AND EQUITY AS SOURCES

The purpose here is to make debt and equity as sources concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Debenture** means A bond backed only by the issuer's general promise to pay, not by a particular asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Retained earnings** means Profit kept in the business instead of being paid out as dividends. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital market** means The market in which long-term money is raised by selling bonds and shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For debt and equity as sources, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

The business meaning matters as much as the arithmetic.

This section builds the idea of debt and equity as sources before moving to application.

Debenture. A bond backed only by the issuer's general promise to pay, not by a particular asset..

The term debenture gives the chapter a precise name for this part of the problem.

Retained earnings. Profit kept in the business instead of being paid out as dividends..

The term retained earnings gives the chapter a precise name for this part of the problem.

Capital market. The market in which long-term money is raised by selling bonds and shares..

The term capital market gives the chapter a precise name for this part of the problem.

The central question in debt and equity as sources is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Richard Price, the 1772 Appeal on the National Debt, and the British sinking fund of 1786 that his compound-interest argument helped to inspire. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

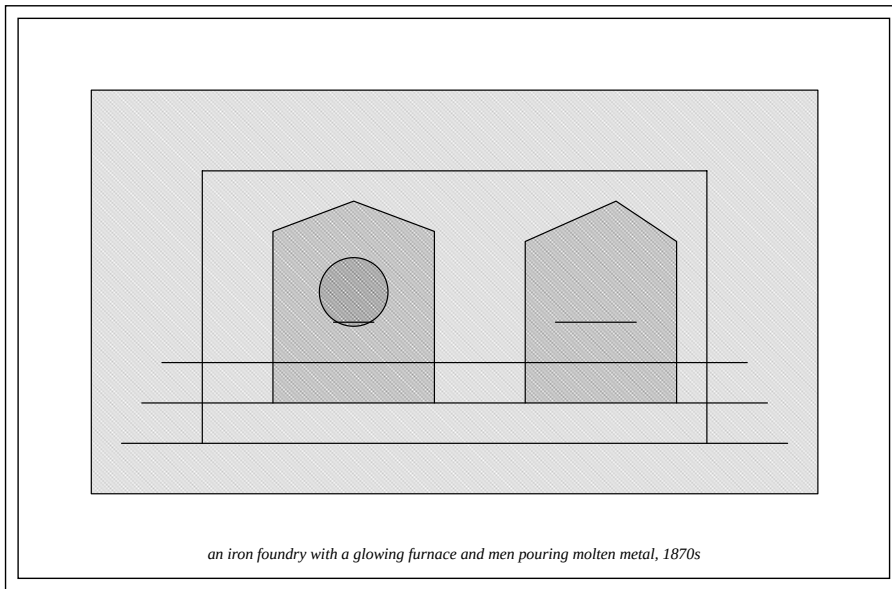
04.T1 explain why a taka today beats a taka next year, with new figures

04.T2 compute simple and compound interest on Tk 50,000 at 12% for four years

04.T3 use a factor table to find a future value for a new rate and period

PAUSE AND CHECK 04.P1 After opportunity cost, state the rule in your own words.

PAUSE AND CHECK 04.P2 After compound interest, state the rule in your own words.



CHAPTER XV

Long-Term Financing and the Capital Market

A practical question about long-term financing and the capital market



This chapter introduces long-term financing and the capital market for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

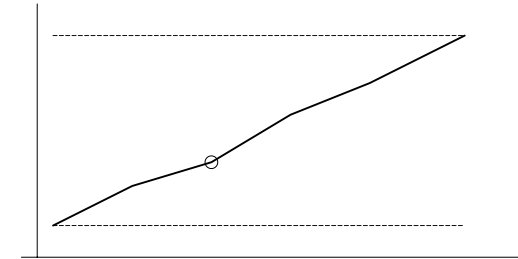
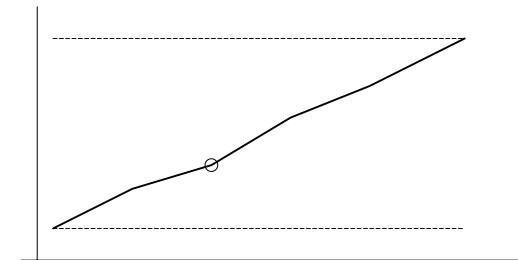
This section builds the idea of discounting a single sum before moving to application.

Present value. What a sum due in the future is worth today..

The term present value gives the chapter a precise name for this part of the problem.

Discounting. Finding present value by removing the interest a sum could earn before it arrives..

The term discounting gives the chapter a precise name for this part of the problem.

FIG. 5.1. *Present Value and Discounting figure 1*FIG. 5.2. *Present Value and Discounting figure 2*

5.1 DISCOUNTING A SINGLE SUM

The purpose here is to make discounting a single sum concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Present value** means What a sum due in the future is worth today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discounting** means Finding present value by removing the interest a sum could earn before it arrives. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

PAUSE AND CHECK 14.P1 After matching principle, state the rule in your own words.

PAUSE AND CHECK 14.P2 After cash conversion cycle, state the rule in your own words.

* * *

5.3 COMPARING STREAMS OF CASH FLOWS

The purpose here is to make comparing streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Cash outflow** means Money going out of a business or from an investor, shown as a negative number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Mixed stream** means A series of cash flows of different sizes in different periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For comparing streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to present value and discounting. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

This section builds the idea of comparing streams of cash flows before moving to application.

Cash outflow. Money going out of a business or from an investor, shown as a negative number..

The term cash outflow gives the chapter a precise name for this part of the problem.

Mixed stream. A series of cash flows of different sizes in different periods..

The term mixed stream gives the chapter a precise name for this part of the problem.

The central question in comparing streams of cash flows is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Accounts payable** means Money a business owes its suppliers for goods already received. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Operating cycle** means The days from buying stock to collecting cash from its sale. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash conversion cycle** means The operating cycle minus the days a business takes to pay its suppliers. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Float** means Money tied up in payments that are sent but not yet usable, such as a cheque in the post. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For cycles, float and estimating needs, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to working capital and short-term financing. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Accounts payable. Money a business owes its suppliers for goods already received..

The term accounts payable gives the chapter a precise name for this part of the problem.

Operating cycle. The days from buying stock to collecting cash from its sale..

The term operating cycle gives the chapter a precise name for this part of the problem.

Cash conversion cycle. The operating cycle minus the days a business takes to pay its suppliers..

The term cash conversion cycle gives the chapter a precise name for this part of the problem.

Float. Money tied up in payments that are sent but not yet usable, such as a cheque in the post..

The term float gives the chapter a precise name for this part of the problem.

The central question in cycles, float and estimating needs is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

IN HISTORY

Irving Fisher, The Theory of Interest (1930), which states the discounting idea in its modern form. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

05.T1 explain discounting as the reverse of compounding using new figures

05.T2 find the present value of a single sum for two discount rates

05.T3 find present value of a mixed stream and compare two streams

PAUSE AND CHECK 05.P1 After present value of a single sum, state the rule in your own words.

PAUSE AND CHECK 05.P2 After the mixed stream, state the rule in your own words.

In this chapter, **Net working capital** means Current assets minus current liabilities; what remains for trading after short debts are paid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Matching principle** means Pay for short-lived needs with short-term money and for lasting needs with long-term money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Short-term financing** means Money borrowed to be repaid within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Intermediate-term financing** means Money borrowed to be repaid in more than one year but not more than about five. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For working capital and how to finance it, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

14.2 SOURCES OF SHORT-TERM FINANCE

The purpose here is to make sources of short-term finance concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Net working capital. Current assets minus current liabilities; what remains for trading after short debts are paid.

The term net working capital gives the chapter a precise name for this part of the problem.

Matching principle. Pay for short-lived needs with short-term money and for lasting needs with long-term money..

The term matching principle gives the chapter a precise name for this part of the problem.

Short-term financing. Money borrowed to be repaid within one year..

The term short-term financing gives the chapter a precise name for this part of the problem.

Intermediate-term financing. Money borrowed to be repaid in more than one year but not more than about five..

The term intermediate-term financing gives the chapter a precise name for this part of the problem.

The central question in working capital and how to finance it is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of sources of short-term finance before moving to application.

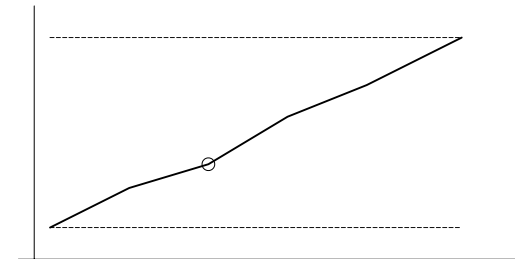


FIG. 6.1. Annuities figure 1

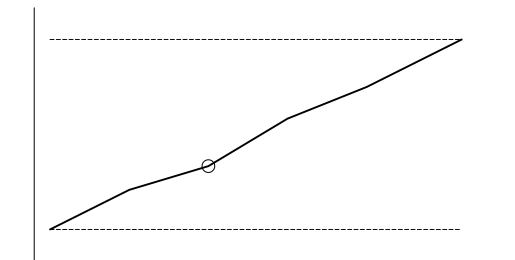


FIG. 6.2. Annuities figure 2

6.1 WHAT AN ANNUITY IS

The purpose here is to make what an annuity is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Annuity** means A series of equal payments made at equal intervals for a fixed number of periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

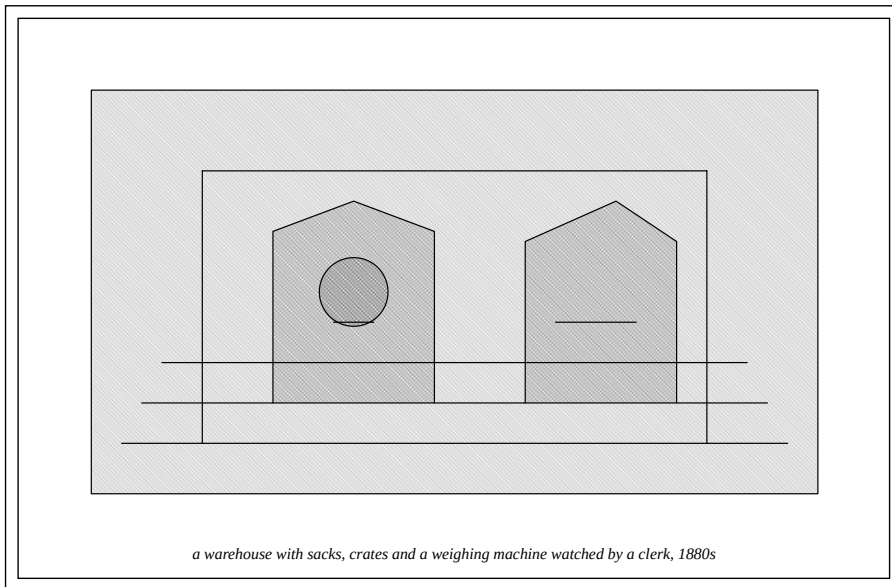
Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

This section builds the idea of what an annuity is before moving to application.

Annuity. A series of equal payments made at equal intervals for a fixed number of periods..

The term annuity gives the chapter a precise name for this part of the problem.

The central question in what an annuity is what changes when the underlying decision changes.



CHAPTER XIV

Working Capital and Short-Term Financing

A practical question about working capital and short-term financing



This chapter introduces working capital and short-term financing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Annuity due. An annuity whose payments fall at the start of each period..

The term annuity due gives the chapter a precise name for this part of the problem.

The central question in present value of an annuity is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of solving for the payment before moving to application.

Instalment. One of a series of regular payments used to repay a loan or build a sum..

The term instalment gives the chapter a precise name for this part of the problem.

The central question in solving for the payment is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

In this chapter, **Annuity due** means An annuity whose payments fall at the start of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For present value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.4 SOLVING FOR THE PAYMENT

The purpose here is to make solving for the payment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Instalment** means One of a series of regular payments used to repay a loan or build a sum. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For solving for the payment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Valuation of Shares is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

The South Sea Company of 1720: its share price rose and collapsed in one year while its earnings stayed small. Valuation theory: John Burr Williams, *The Theory of Investment Value* (1938) and Myron Gordon (1959). The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

SOURCES

Chapter 12, 'Valuation of Bonds', section 12.1 'The idea of valuation' — the principle used; Chapter 11, 'Required Return, CAPM and Market Efficiency', section 11.3 'The capital asset pricing model' — the required return; Chapter 15, 'Long-Term Financing and the Capital Market', section 15.2 'Debt and equity as sources' — stock as a source of money..

REVIEW QUESTIONS

13.T1 say how common stock differs from a bond using a new firm

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

The chapter points back to related ideas elsewhere in the book.

A short restatement keeps the financial meaning attached to the calculation.

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each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Annuities is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Johan de Witt, 1671 report to the States of Holland on valuing life annuities for the Dutch Republic. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

06.T1 say what makes a series of payments an annuity and name its two timing forms

06.T2 find the future value of an annuity for several rate and period pairs as ordinary and as due

06.T3 find the present value of a new instalment plan

06.T4 compute the yearly deposit needed for a stated goal

PAUSE AND CHECK 06.P1 After annuity types, state the rule in your own words.

PAUSE AND CHECK 06.P2 After future value of an ordinary annuity, state the rule in your own words.

* * *

13.2 VALUING PREFERRED STOCK

The purpose here is to make valuing preferred stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital gain** means The profit from selling an asset for more than its purchase price. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Perpetuity** means A series of equal payments that continues for ever. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing preferred stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

13.3 VALUING COMMON STOCK

The purpose here is to make valuing common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend valuation model** means A way of valuing a share as the present value of the dividends it is expected to pay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of valuing preferred stock before moving to application.

Capital gain. The profit from selling an asset for more than its purchase price..

The term capital gain gives the chapter a precise name for this part of the problem.

Perpetuity. A series of equal payments that continues for ever..

The term perpetuity gives the chapter a precise name for this part of the problem.

The central question in valuing preferred stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of valuing common stock before moving to application.

Dividend valuation model. A way of valuing a share as the present value of the dividends it is expected to pay..

The term dividend valuation model gives the chapter a precise name for this part of the problem.

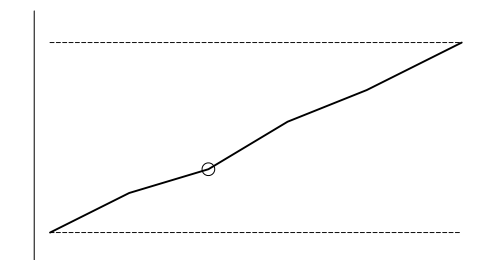


FIG. 7.1. Capital Budgeting: Process and Cash Flows figure 1

7.1 WHY FIRMS BUDGET FOR CAPITAL

The purpose here is to make why firms budget for capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital budgeting** means Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital expenditure** means Spending on an asset that will help a business for more than one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Independent project** means A project that can be accepted or rejected without affecting any other project. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Mutually exclusive projects** means Projects of which only one can be chosen, because choosing one rules out the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

This section builds the idea of why firms budget for capital before moving to application.

Capital budgeting. Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring..

The term capital budgeting gives the chapter a precise name for this part of the problem.

Capital expenditure. Spending on an asset that will help a business for more than one year..

The term capital expenditure gives the chapter a precise name for this part of the problem.

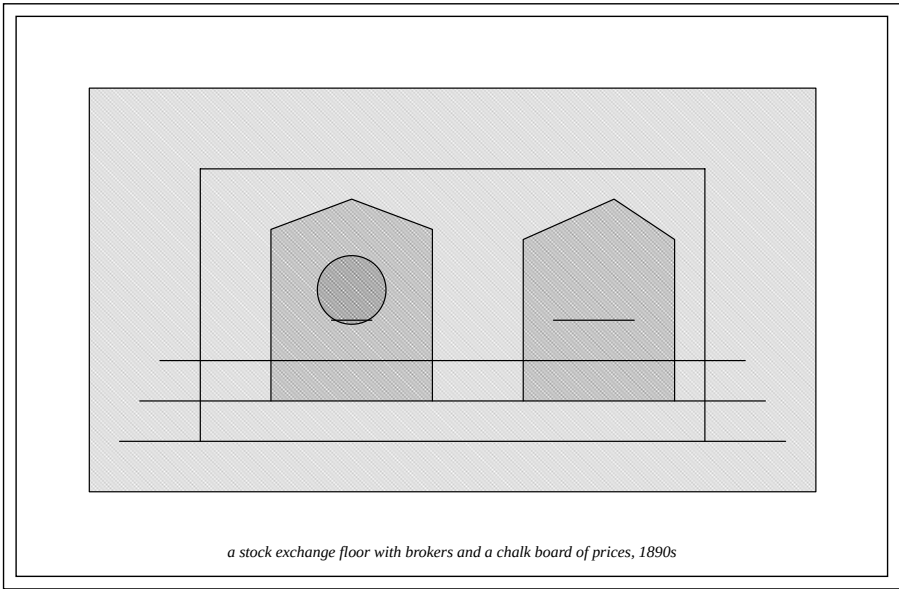
Independent project. A project that can be accepted or rejected without affecting any other project..

The term independent project gives the chapter a precise name for this part of the problem.

Mutually exclusive projects. Projects of which only one can be chosen, because choosing one rules out the other..

The term mutually exclusive projects gives the chapter a precise name for this part of the problem.

The central question in why firms budget for capital is what changes when the underlying decision changes.



a stock exchange floor with brokers and a chalk board of prices, 1890s

CHAPTER XIII

Valuation of Shares

A practical question about valuation of shares



This chapter introduces valuation of shares for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

The business meaning matters as much as the arithmetic.

This section builds the idea of cash flows from an investment before moving to application.

Terminal cash flow. The cash received or paid when a project ends, such as the sale of scrap..

The term terminal cash flow gives the chapter a precise name for this part of the problem.

Depreciation. The share of an asset's cost charged against income each year as the asset wears out..

The term depreciation gives the chapter a precise name for this part of the problem.

The central question in cash flows from an investment is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the steps of the process, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

7.3 CASH FLOWS FROM AN INVESTMENT

The purpose here is to make cash flows from an investment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Terminal cash flow** means The cash received or paid when a project ends, such as the sale of scrap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Depreciation** means The share of an asset's cost charged against income each year as the asset wears out. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For cash flows from an investment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to capital budgeting: process and cash flows. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep

finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.4 YIELD AND PRICE

The purpose here is to make yield and price concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For yield and price, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to valuation of bonds. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

This section builds the idea of yield and price before moving to application.

The central question in yield and price is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

07.T4 find the operating cash flow from a new set of sales, costs, depreciation and tax

PAUSE AND CHECK 07.P1 After the steps, state the rule in your own words.

PAUSE AND CHECK 07.P2 After relevant cash flows, state the rule in your own words.

In this chapter, **Bond** means A loan made to a company or government, repaid on a set date, with interest paid on the way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Face value** means The amount a bond repays at its end, for example Tk 1,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the idea of valuation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.2 WHAT A BOND IS

The purpose here is to make what a bond is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Coupon rate** means The yearly interest of a bond, as a percentage of its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Maturity** means The date on which a bond or loan must be repaid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Yield to maturity** means The yearly return an investor earns by buying a bond at today's price and holding it to maturity. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Bond. A loan made to a company or government, repaid on a set date, with interest paid on the way..

The term bond gives the chapter a precise name for this part of the problem.

Face value. The amount a bond repays at its end, for example Tk 1,000..

The term face value gives the chapter a precise name for this part of the problem.

The central question in the idea of valuation is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of what a bond is before moving to application.

Coupon rate. The yearly interest of a bond, as a percentage of its face value..

The term coupon rate gives the chapter a precise name for this part of the problem.

Maturity. The date on which a bond or loan must be repaid..

The term maturity gives the chapter a precise name for this part of the problem.

Yield to maturity. The yearly return an investor earns by buying a bond at today's price and holding it to maturity..

The term yield to maturity gives the chapter a precise name for this part of the problem.

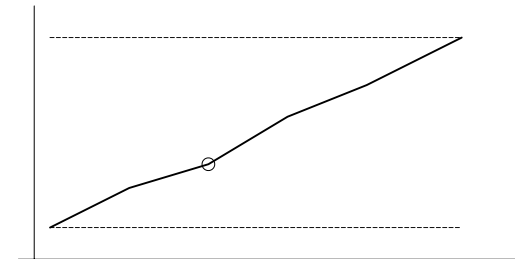


FIG. 8.1. Evaluating Projects: Payback and Net Present Value figure 1

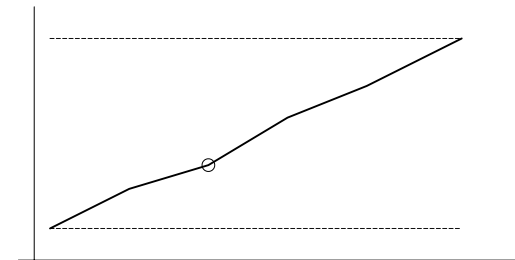


FIG. 8.2. Evaluating Projects: Payback and Net Present Value figure 2

8.1 PAYBACK PERIOD

The purpose here is to make payback period concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Payback period** means The time a project takes to return its initial investment from its cash flows. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discounted payback period** means The payback time when each cash flow is first discounted to present value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of payback period before moving to application.

Payback period. The time a project takes to return its initial investment from its cash flows..

The term payback period gives the chapter a precise name for this part of the problem.

Discounted payback period. The payback time when each cash flow is first discounted to present value..

The term discounted payback period gives the chapter a precise name for this part of the problem.

* * *

8.3 OTHER MEASURES OF A PROJECT

The purpose here is to make other measures of a project concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profitability index** means Present value of future cash flows divided by the initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For other measures of a project, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

8.4 CHOOSING AMONG METHODS

The purpose here is to make choosing among methods concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For choosing among methods, keep the decision in view. Ask what the business is trying

This section builds the idea of other measures of a project before moving to application.

Profitability index. Present value of future cash flows divided by the initial investment..

The term profitability index gives the chapter a precise name for this part of the problem.

The central question in other measures of a project is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of choosing among methods before moving to application.

The central question in choosing among methods is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

CHAPTER XII

Valuation of Bonds

A practical question about valuation of bonds



This chapter introduces valuation of bonds for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Required Return, CAPM and Market Efficiency is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs

A short restatement keeps the financial meaning attached to the calculation.

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A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A short restatement keeps the financial meaning attached to the calculation.

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later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.4 EFFICIENT MARKETS

The purpose here is to make efficient markets concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital asset pricing model** means A formula for required return: risk-free rate plus beta times the market risk premium. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Security market line** means The graph of the capital asset pricing model, with beta on one axis and required return on the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Efficient market** means A market in which prices already reflect the available information. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For efficient markets, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to required return, capm and market efficiency. Start by listing the known amounts and their dates. Separate cash inflows from

This section builds the idea of efficient markets before moving to application.

Capital asset pricing model. A formula for required return: risk-free rate plus beta times the market risk premium..

The term capital asset pricing model gives the chapter a precise name for this part of the problem.

Security market line. The graph of the capital asset pricing model, with beta on one axis and required return on the other..

The term security market line gives the chapter a precise name for this part of the problem.

Efficient market. A market in which prices already reflect the available information..

The term efficient market gives the chapter a precise name for this part of the problem.

The central question in efficient markets is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

WHERE PEOPLE GO WRONG

Most errors arise from mixing terms, timing, signs, or decision rules.

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

A historical case shows how the chapter's financial problem appeared in practice.

The Channel Tunnel, opened 6 May 1994, whose traffic forecasts and costs differed greatly from plan. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

08.T1 compute payback and net present value for new project data and state the decision rule for each

08.T2 say what the internal rate of return means in plain words

08.T3 rank three new projects by profitability index

PAUSE AND CHECK 08.P1 After payback, state the rule in your own words.

PAUSE AND CHECK 08.P2 After net present value, state the rule in your own words.

In this chapter, **Risk premium** means The extra return demanded for taking risk instead of a risk-free investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For risk premium and required return, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.2 DIVERSIFICATION AND BETA

The purpose here is to make diversification and beta concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Portfolio** means A collection of investments held together by one investor. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Diversification** means Spreading money across many investments so that one failure does less damage. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Systematic risk** means Risk that affects all investments, such as a recession; it cannot be diversified away. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Risk premium. The extra return demanded for taking risk instead of a risk-free investment..

The term risk premium gives the chapter a precise name for this part of the problem.

The central question in risk premium and required return is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of diversification and beta before moving to application.

Portfolio. A collection of investments held together by one investor..

The term portfolio gives the chapter a precise name for this part of the problem.

Diversification. Spreading money across many investments so that one failure does less damage..

The term diversification gives the chapter a precise name for this part of the problem.

Systematic risk. Risk that affects all investments, such as a recession; it cannot be diversified away..

The term systematic risk gives the chapter a precise name for this part of the problem.

The central question in diversification and beta is what changes when the underlying decision changes.

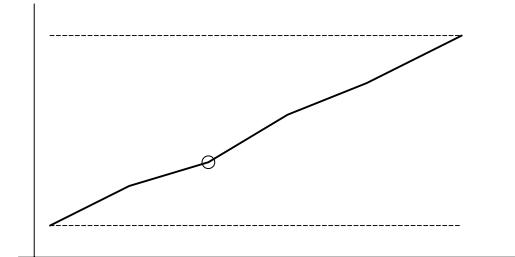


FIG. 9.1. Risk and Return figure 1

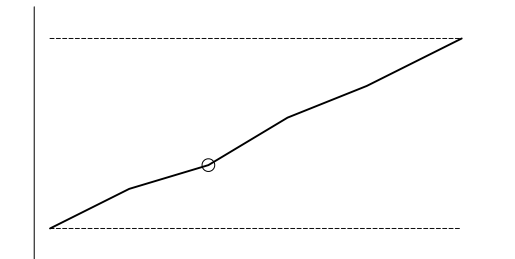


FIG. 9.2. Risk and Return figure 2

9.1 RETURN AND RISK

The purpose here is to make return and risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Return** means The gain or loss from an investment, including income received and any change in value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Rate of return** means The return in one period divided by the sum invested, written as a percentage. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

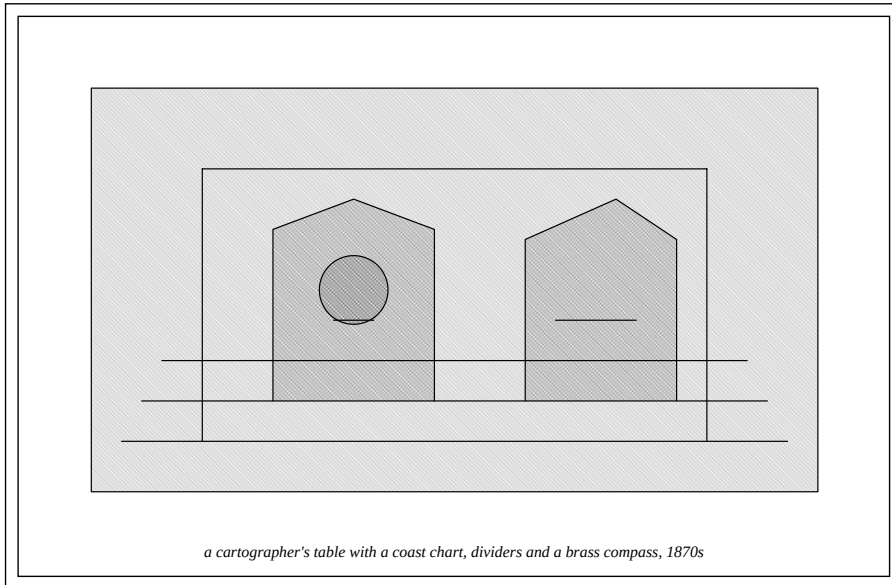
This section builds the idea of return and risk before moving to application.

Return. The gain or loss from an investment, including income received and any change in value..

The term return gives the chapter a precise name for this part of the problem.

Rate of return. The return in one period divided by the sum invested, written as a percentage..

The term rate of return gives the chapter a precise name for this part of the problem.



CHAPTER XI

Required Return, CAPM and Market Efficiency

A practical question about required return, capm and market efficiency



This chapter introduces required return, capm and market efficiency for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Expected value. The average of possible results, each weighted by its probability..

The term expected value gives the chapter a precise name for this part of the problem.

Variance. The probability-weighted average of the squared gaps between each result and the expected value..

The term variance gives the chapter a precise name for this part of the problem.

The central question in describing uncertainty is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

In this chapter, **Expected value** means The average of possible results, each weighted by its probability. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Variance** means The probability-weighted average of the squared gaps between each result and the expected value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For describing uncertainty, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

9.3 MEASURING RISK

This section builds the idea of measuring risk before moving to application.

Standard deviation. The square root of the variance; it shows how far results typically stray from the expected value..

The term standard deviation gives the chapter a precise name for this part of the problem.

Coefficient of variation. Standard deviation divided by expected value, giving risk per unit of expected return..

The term coefficient of variation gives the chapter a precise name for this part of the problem.

The central question in measuring risk is what changes when the underlying decision changes.

The purpose here is to make measuring risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Standard deviation** means The square root of the variance; it shows how far results typically stray from the expected value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Coefficient of variation** means Standard deviation divided by expected value, giving risk per unit of expected return. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Leverage is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

A short restatement keeps the financial meaning attached to the calculation.

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The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A short restatement keeps the financial meaning attached to the calculation.

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larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

10.4 TOTAL LEVERAGE AND RISK

The purpose here is to make total leverage and risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Earnings per share** means Earnings left for owners of common stock divided by the number of shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Preferred stock** means Shares that pay a fixed dividend, which must be paid before any dividend on common stock. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For total leverage and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to leverage. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000.

This section builds the idea of total leverage and risk before moving to application.

Earnings per share. Earnings left for owners of common stock divided by the number of shares..

The term earnings per share gives the chapter a precise name for this part of the problem.

Preferred stock. Shares that pay a fixed dividend, which must be paid before any dividend on common stock..

The term preferred stock gives the chapter a precise name for this part of the problem.

The central question in total leverage and risk is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

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SUMMARY

The central purpose of Risk and Return is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Daniel Bernoulli, 1738 paper on measuring risk, which showed that people judge a gamble by its meaning to them as well as by its average. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

09.T1 define risk and say how business risk differs from financial risk, using a new firm

09.T2 compute the expected value, then the variance, the standard deviation and the coefficient of variation, for a new two-asset table and state which is riskier per unit of return

09.T3 explain standard deviation in plain words

In this chapter, **Variable cost** means A cost that rises and falls with the number of units sold, such as the price paid for each book. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For costs and the break-even point, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

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10.2 OPERATING LEVERAGE

The purpose here is to make operating leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Contribution margin** means Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Break-even point** means The sales volume at which total revenue equals total cost, so profit is zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **EBIT** means Earnings before interest and tax: sales minus operating costs. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Variable cost. A cost that rises and falls with the number of units sold, such as the price paid for each book..

The term variable cost gives the chapter a precise name for this part of the problem.

The central question in costs and the break-even point is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of operating leverage before moving to application.

Contribution margin. Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit..

The term contribution margin gives the chapter a precise name for this part of the problem.

Break-even point. The sales volume at which total revenue equals total cost, so profit is zero..

The term break-even point gives the chapter a precise name for this part of the problem.

EBIT. Earnings before interest and tax: sales minus operating costs..

The term ebit gives the chapter a precise name for this part of the problem.

The central question in operating leverage is what changes when the underlying decision changes.